

Question 1

Parameters

- $G = (V, E)$: A directed graph with vertex set V and edge set E .
- K : The number of commodities, indexed by $k = 1, \dots, K$.
- $s_k \in V$: Source node of commodity k .
- $t_k \in V$: Sink node of commodity k .
- $d_k > 0$: Demand for commodity k , the total amount of commodity k that must be shipped from s_k to t_k .
- $u_{ij} \geq 0$: Capacity of arc $(i, j) \in E$. This capacity is shared by all commodities flowing through (i, j) .
- $c_{ij}^k \geq 0$: Per-unit cost of sending commodity k along arc $(i, j) \in E$. (Costs may vary by commodity and arc.)

Decision Variables

- $x_{ij}^k \geq 0$: The flow of commodity k on arc (i, j) . That is, x_{ij}^k is how many units of commodity k are shipped through (i, j) .

Objective Function

We want to minimize the total shipping cost for all commodities on all arcs:

$$\text{Minimize} \quad \sum_{(i,j) \in E} \sum_{k=1}^K c_{ij}^k x_{ij}^k.$$

Constraints

1. Flow-Balance Constraints. For each commodity k and each node $v \in V$, the difference between the total incoming flow and the total outgoing flow must equal the net supply at node v . Specifically:

$$\sum_{(u,v) \in E} x_{uv}^k - \sum_{(v,w) \in E} x_{vw}^k = \begin{cases} d_k, & \text{if } v = s_k, \\ -d_k, & \text{if } v = t_k, \\ 0, & \text{otherwise.} \end{cases}$$

$$\forall v \in V, \quad \forall k = 1, \dots, K.$$

This ensures each commodity's flow is conserved: - at its source s_k , we “add” a total of d_k units, - at its sink t_k , we “extract” d_k units, - and at intermediate nodes, net flow is zero.

2. Capacity Constraints. Since each arc (i, j) has capacity u_{ij} that is shared by *all* commodities,

$$\sum_{k=1}^K x_{ij}^k \leq u_{ij}, \quad \forall (i, j) \in E.$$

No more than u_{ij} units of combined flow (across all commodities) can go through (i, j) .

3. Nonnegativity.

$$x_{ij}^k \geq 0, \quad \forall (i, j) \in E, \quad \forall k = 1, \dots, K.$$

Question 2

Sets

- Let $S = \{1, 2, \dots, 8\}$ be the set of stations.

- Let

$$K = \{(i, j) \mid i, j \in S, i \neq j\}$$

be the set of all origin-destination (OD) pairs.

- Let R be the set of all enumerated routes. Each route $r \in R$ is an ordered sequence of stations:

$$r = (s_1^r, s_2^r, \dots, s_{L_r}^r),$$

where L_r is the number of stops on route r . The route is monotonic (either strictly ascending or strictly descending).

Parameters

- **Passenger Demand:** For each $(i, j) \in K$, let

d_{ij} be the number of passengers from station i to j .

- **Travel Time:** For any $i, j \in S$, let

t_{ij} be the travel time (in minutes) from station i to j .

- **Route-Specific Parameters:**

- For route r and any two stations i, j that appear in r in the correct order (i.e., i appears before j), define the cumulative travel time

$$T_{ij}^r = \text{cumulative travel time from } i \text{ to } j \text{ along route } r.$$

If the route does not serve the OD pair (i, j) (either one of the stations is missing or they appear in reverse order), then

$$T_{ij}^r = 0.$$

- For a route r with L_r stops, the route cost is given by

$$c_r = \text{fixed_cost} + \text{station_cost} \times L_r,$$

where $\text{fixed_cost} = 1000$ and $\text{station_cost} = 100$.

- **Operational Parameters:**

- Bus capacity: $Q = 150$.
- Maximum buses per route: $M = 5$.
- Total travel time cap: $T_{\text{cap}} = 391890$.

Decision Variables

For each route $r \in R$ and each OD pair $(i, j) \in K$:

- $y_r \in \{0, 1\}$: equals 1 if route r is activated (used), and 0 otherwise.
- $b_r \in \{0, 1, \dots, M\}$: the number of buses assigned to route r .
- $x_{ij}^r \geq 0$: the number of passengers for OD pair (i, j) assigned to route r .

Objective Function

Minimize the total cost of activated routes:

$$\min \sum_{r \in R} c_r y_r.$$

Constraints

1. **Demand Satisfaction:** For every OD pair $(i, j) \in K$,

$$\sum_{r \in R} x_{ij}^r = d_{ij}.$$

2. **Capacity Constraint:** For each route $r \in R$,

$$\sum_{(i,j) \in K} x_{ij}^r \leq Q b_r.$$

3. **Bus-Route Linking:** To ensure that buses are only assigned if the route is activated,

$$b_r \leq M y_r \quad \forall r \in R.$$

4. **Total Travel Time Cap:** The total travel time used by all passengers must not exceed T_{cap} :

$$\sum_{r \in R} \sum_{(i,j) \in K} T_{ij}^r x_{ij}^r \leq T_{\text{cap}}.$$

5. **Route Feasibility:** If a route r does not serve an OD pair (i, j) (i.e., $T_{ij}^r = 0$), then

$$x_{ij}^r = 0 \quad \forall (i, j) \in K, \forall r \in R.$$